

KSR Akshara Academy

Tiruchengode

Holiday Assignment

Grade :IX

KSR Akshara Academy

Tiruchengode

Dear Parents,

- KSR Akshara's **ADMISSION** for the year **2025- 26** is going on for the grades **Kg to XI**. Kindly share it with your friends, neighbours and relatives to join us to unlock their child's potential and be with our best team.
- **Parent referral scheme** is ongoing. Kindly refer your friends, neighbours and relatives to join at Akshara and shape their child's future.
- **Summer Vacation** starts from **April 18, 2025 to June 04, 2025**.
- The **school reopens** after Summer Vacation for the grades **SKg to IX on June 05, 2025** (Thursday) and **Pre Kg & JKg on June 09, 2025** (Monday)
- Kindly ensure that your child follows a **routine** during holidays.
- Give them more time to learn about your family members. Share them your past history and the developmental growth you had in your life and also add on the belief and expectations you have with your child etc.
- Please spend your precious time with your child every day. Take them out in the evening times and have dine with your family together during their holidays.
- Kindly guide and monitor your child for the completion of their Holiday Assignment. The completed Assignments should be submitted on **June 05, 2025** without any excuses.
- Based on your child's submission and presentation, the students will get **Reward Points** added on for their **Internal Assessment** of their respective subjects and that will be included for their **Periodic Assessment – I.**
- Also ensure that your child follows the following **Summer Tips**.
  1. Should stay active with a proper schedule.
  2. Make them follow our Wellness Calendar regularly.
  3. Let them read more books and newspaper.
  4. Let them learn a new skill or hobby. Encourage and give them opportunity to practice a musical instrument.
  5. Engage them in creative activities like drawing, painting, writing etc.
  6. It would be more beneficial if they write a paragraph in English and Tamil neatly every dayto improvise their handwriting.
  7. Enroll in summer classes and make them learn something new.
  8. Remind them to drink plenty of water throughout the day and stay hydrated.
  9. Let them drink more tender coconut or fresh juices and avoid spicy food.
  10. Play outdoors only in the evening.
  11. Let them follow healthy eating habits.
  12. **Limit their screen time** and make them stay away from online strangers.
  13. Use strong passwords and be cautious with the online content what they use.
  14. Avoid encouraging them to be in touch with social media chatting and unnecessary postings.
  15. Let them go to bed early.

Holiday Activity

ENGLISH

1. **Project Title**: "Exploring Literature Through Alternate Endings"
Objective: To enhance creative writing skills, critical thinking, and understanding of narrative structures by rewriting the ending of a story/novel from the English syllabus.

Task:
Choose a Story/Novel: Pick one of the following from your syllabus:

- The Fun They Had (Isaac Asimov)
- The Lost Child (Mulk Raj Anand)
- The Adventures of Toto (Ruskin Bond)
- The Last Leaf (O. Henry) – (If included in supplementary reader)

Rewrite the Ending:

- Change the climax or conclusion of the story in a creative way.
- Maintain the original theme or give it a new twist.
- Word limit: 300-400 words.



Justify Your Ending (100 words):

- Explain why you changed the ending.
- How does your version impact the message of the story?

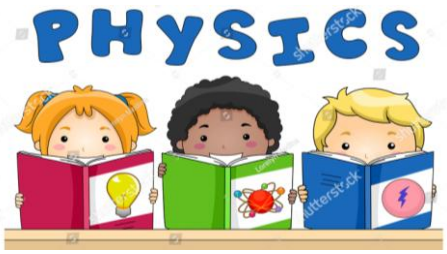
Presentation:

- Write/type your alternate ending neatly.
- Add illustrations (optional but encouraged).

Submission Guidelines:

- Handwritten or printed (A4 sheet).
- Include a title, your name, and class.
- Decorate the project with relevant drawings or digital designs.

PHYSICS



1) Activity-Based :

1. Ball Rolling Experiment:

- Take a ball, roll it on a smooth floor and measure:
- Time taken to travel a fixed distance using a stopwatch.
- Calculate the average velocity (assume straight path = displacement).
- Observe if the ball's speed decreases over time. Why does it happen?

2. Measuring Distance and Displacement around a Playground

Walk along the boundary of a nearby playground, starting and ending at the Same point. Use a measuring tape or fitness app to note the total path length.

Question:

a) What is the total distance you covered?

b) What is your displacement?

c) Explain why the displacement is different from the distance in this case.

3. Measuring Distance and Displacement on a Zig-Zag Path

Create a zig-zag path using markers or stones in an open area (like your garden or terrace). Walk from start to finish, following the zig-zag path.

Question:

a) Measure and write the total distance you walked.

b) Measure the straight-line distance from your start point to your endpoint (displacement).

c) Which one is greater — distance or displacement? Why?

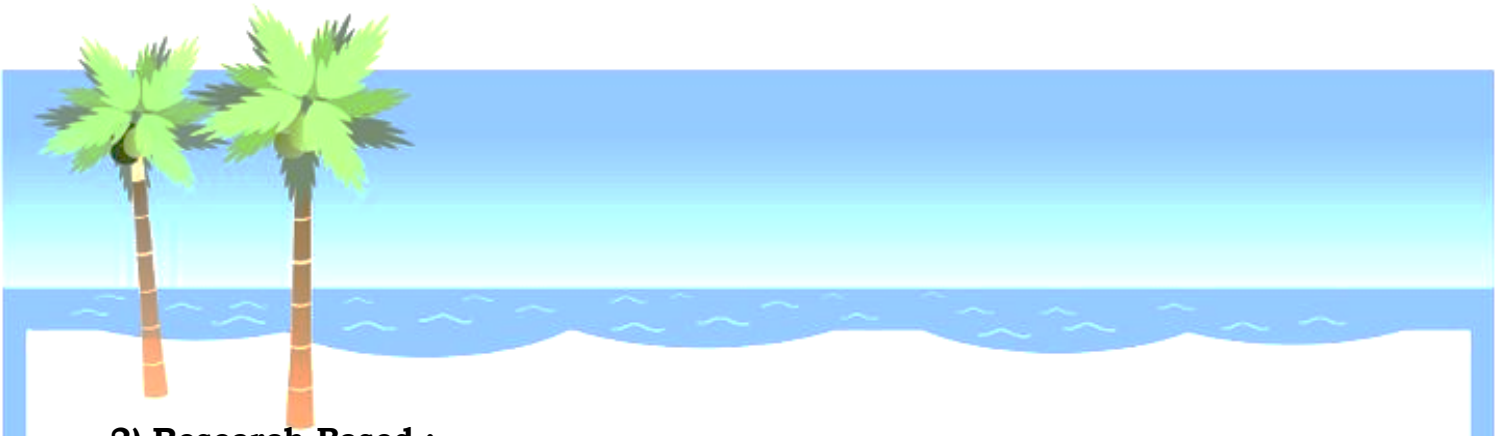
4. Staircase Climbing Challenge

Climb from the ground floor to the 3rd floor in your house or apartment. Measure the total path using step count × height of one step, and use a measuring tape to measure the straight-line distance (displacement).

a) Write the total distance covered along the stairs.

b) Write the displacement (from ground to 3rd floor).

c) Which is larger distance or displacement? Why?



2) Research-Based :

Research and write a short note (about 100–150 words) on the role of motion in daily life

3) Question Paper-Based:

Solve the following questions:

a) A car moves with a speed of 40 km/h for 2 hours and then 60 km/h for 3 hours. Find the total distance covered and the average speed.

b) An object starts from rest and accelerates uniformly at 5 m/s^2 for 8 seconds. Find:

i) Final velocity

ii) Distance covered in 8 seconds

c) Draw a velocity-time graph for an object moving with uniform acceleration.

4) Problem-Based:

1. Numerical Problem:

A car travels 150 km at a speed of 50 km/h and then 120 km at a speed of 60 km/h. Calculate:

a) The total time taken for the journey.

b) The average speed of the car for the whole trip.

2. Numerical Problem:

An object is moving with an initial velocity of 5 m/s and it accelerates uniformly at 3 m/s^2 for 6 seconds. Calculate:

a) The final velocity.

b) The distance covered in these 6 seconds.

3. Numerical Problem:

A train starting from rest attains a speed of 72 km/h in 5 minutes. Assuming uniform acceleration, calculate:

a) The acceleration of the train.

b) The distance travelled during this time.

4. Conceptual Problem: A student plots a distance-time graph for a moving object and the graph is a straight line but parallel to the time-axis. What can you conclude about the motion of the object? Explain

Book Reading

Objective

To read and review two selected books from a given list, developing critical Thinking and literary analysis skills.



1. Choose two books from the provided list of five books.
2. Ensure the selected books align with your interests.
3. Read the selected books thoroughly.
4. Take notes on:

- Book Title
- No of pages
- Author
- Pages Read
- Characters and Write a short note about the character whom you like most.
- Summary

Books name

1. The Beyonders.
2. Wonders
3. Diary of a Wimpy Kid (Any one from series)
4. Charlie the Chocolate
5. The Adventure of Tom Sawyer

BIOLOGY



1. Model Making – 3D Cell Model

Create a 3D model of a plant or animal cell using household materials:

Use clay, cardboard, beads, buttons, or food items (e.g., jelly for cytoplasm, lentils for mitochondria).

Label all organelles with brief functions.

Submission: Click photos/video explaining your model.

SOCIAL

Geography



Title: Ch:1 India, Size and Location

News Scrapbook:

Collect 3 news clippings or headlines (from newspaper/internet) that mention India's role in global trade, sea routes, or border relations. Paste them and write 2-3 lines on how India's location plays a role.

2. "Time Zone* Find out and list the local times in any 4 countries (e.g., Japan, UK, USA, Australia) when it is 12:00 noon in India. Use this to explain why we need a Standard Time.

History

Create an activity book which includes all the pictures of French revolution from the textbook. (Only pictures to be downloaded and to be stuck)

Include any 5 interesting info about Napoleon Bonaparte.





MATHEMATICS



NUMBER SYSTEM

LearningOutcomes

Studentsareableto

Represent realnumbersonthenumberline

1. ACTIVITY-SQUARE ROOT SPIRAL

Instructions

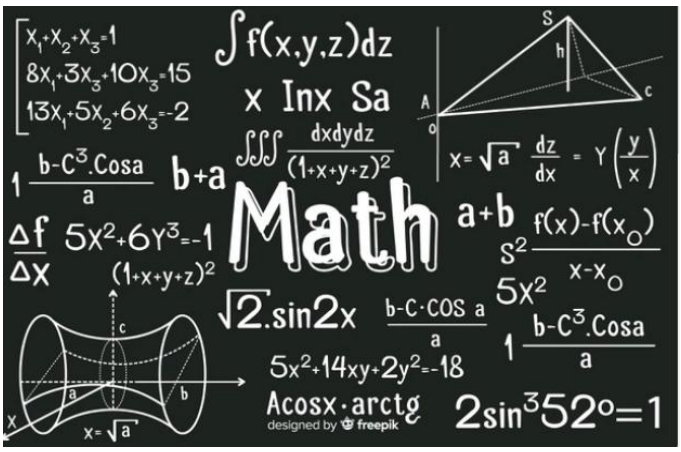
oRepresent$\sqrt{2},\sqrt{3},\sqrt{4},\sqrt{5},\sqrt{6},\sqrt{7}$..... *in a continuous manner*.

- A sample image for the spiral has been attached; use your creativity to create the square root spiral.

Project – KNOW ABOUT STORY OF PI (π)

The number pi (π) is one of the most intriguing and significant constants in mathematics.

Make a project in A4 size sheet in a creative manner about the story of pi which emphasizes the historical development and wide-ranging significance of pi in Mathematics and beyond.



Worksheet

1. Write two rational numbers between: a. √2 and √3 b. $\frac{-2}{5}$ and $\frac{1}{2}$
2. Write two irrational numbers between: a. √3 and √4. b. 2.2 and 2.3.
3. Represent $\sqrt{10}, \sqrt{17}, \sqrt{9.5}$ on a number line.
4. Arrange in ascending order $\sqrt[8]{90}, \sqrt[4]{10}, \sqrt{6}$.
5. Rationalize the denominator of each of the following :

   $a. \frac{1}{1+\sqrt{2}-\sqrt{3}}$ $b. \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}+\sqrt{2}-\sqrt{5}}$
6. Express the following in the form of rational number:

   $a. 2.\overline{34}$ b. $5.4\bar{5}$ c. $5.\bar{2} + 0.4\bar{5}$
7. If $x = \frac{\sqrt{2}-1}{\sqrt{2}+1}$ and $y = \frac{\sqrt{2}+1}{\sqrt{2}-1}$, find the value of x^2+y^2+xy.
8. If $\frac{\sqrt{11}-\sqrt{7}}{\sqrt{11}+\sqrt{7}} = x - y\sqrt{77}$, find the values of x and y where x and y are rational numbers.
9. Simplify : $\left[\left[5\left\{\left(\frac{1}{8}\right)^{\frac{-1}{3}} + \left(\frac{1}{27}\right)^{\frac{-1}{3}}\right\}\right]^2\right]^{\frac{1}{4}}$
10. If $a - 8\sqrt{5}b = \frac{8+\sqrt{5}}{8-\sqrt{5}} - \frac{8-\sqrt{5}}{8+\sqrt{5}}$, find a and b.

